The FIFO linking theorem: affine response and the cut-space frontier

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August 2026

Abstract

This note attacks the general linking theorem underlying the ECHO-FIFO+dummy Gold-quadric realizer. A finite graph is played by opening vertices into a FIFO queue and later closing the front. A close flips a bit when the front has odd degree into the untouched set. The target theorem says that one isolated dummy lets the even player force even total flip parity from either seat, for every graph.

The theorem is not proved here. The contribution is an exact algebraic reduction that replaces the failed local partner/debt induction by a global affine-flow target. Every play has a vector-valued score: its graph of disjoint activity intervals. For every fixed attacker strategy, the linking theorem is equivalent to saying that zero lies in the $\mathbb{F}_2$-affine hull of all response scores. At a real FIFO front f, the edge space splits canonically as

$$E(K_L) = \mathrm{Cut}(K_L) \oplus E(K_{L\setminus\{f\}}),$$

so openings before f closes occupy one cut-space layer and all later play occupies the smaller edge-space layer. The quotient is the classical two-graph coboundary chart: changing the front changes the affine-simplex origin, and successive changes obey an absorption law. This gives a canonical filtration, but not a branchwise induction: an exact $P_4 + d$ state has defender children whose projected continuation affine hulls are disjoint.

Positive results accompany the reduction. The vector live-star identity proves the affine formulation, a parity-cell pairing theorem strictly extends the earlier twin-pair solution, and the cut moment at the initial front contracts affinely. A second handshake refinement colours a vertex by its degree parity and its parity of odd-degree neighbours: every colour class is even on an even-order graph, and a same-colour pair balances all four joint degree/incidence fibres. A two-switch tail lemma closes the first genuine adaptive-pairing obstruction. Exact witnesses show why neither same-degree pairing, greedy FIFO rotation, a bare symplectic quotient, nor a uniformly bounded triangular fan closes the general case. Scalar force sets give a least-counterexample odd-routing lemma, while a stronger finite strategy has an exact ranked repair-forest recursion: consecutive queue matchings may extend, delete their front pair, or flip phase along an alternating path. That flip carries a genuine two-graph curvature term. Tetrahedral coherence controls its label, but attacker pruning destroys the constant-coefficient simplex contraction. The remaining obligation is simultaneous cancellation of the correlated cut and continuation moments by a strategy-relative chain. The note therefore advances the proof frontier without promoting the complete finite census through eight real vertices to a proof.

1 The reduced game

Let $G = (R, E)$ be a finite simple graph on the *real* vertices, and adjoin one isolated vertex d. A state is (U, Q, ko), where U is the set of untouched vertices, Q is an ordered FIFO queue of open vertices, and ko records the one-move prohibition after a vertex is opened onto an empty queue. A legal move is

1. OPEN(x) for any $x \in U$, moving x to the back of Q; or
2. CLOSE(f) for the front f of Q, unless ko is set.

A close of f scores

$$\deg_U(f) \pmod 2.$$

If no move is legal, the forced pass only clears ko. This can occur only after U is empty, when no later close can score. The players have opposite targets for the parity of the total score.

[PROVED] Every vertex opens and closes exactly once. FIFO makes the opening and closing orders identical. Consequently the activity intervals $I_x = [o(x), c(x)]$ never properly nest. For an edge xy with $o(x) < o(y)$, the edge contributes to the flip score precisely when $c(x) < o(y)$, i.e. precisely when I_x and I_y are disjoint. Thus the total score is

$$F_G(h) = \sum_{xy \in E(G)} [I_x \cap I_y = \varnothing] \pmod 2. \tag{1}$$

The proper-interval vocabulary is standard; the adversarial parity problem is not a standard queue-layout problem, because the queue stores vertices and the play itself constructs their intervals.

2 The vector score

Write

$$\mathcal{E}_R = \mathbb{F}_2^{\binom{R}{2}}$$

for the vector space whose coordinates are unordered pairs of real vertices. For a completed history h, define its *disjointness vector*

$$D(h) = \sum_{\{x,y\} \subseteq R} [I_x \cap I_y = \varnothing], xy \in \mathcal{E}_R.$$

Pairing $D(h)$ with the adjacency vector A_G recovers (1).

At any moment let $L = U \cup Q$ be the live set. For a real $x \in L$, put

$$s_L(x) = \sum_{y \in (L \cap R) \setminus \{x\}} xy,$$

and set $s_L(d) = 0$. These are the vertex stars, or simplicial coboundaries, of the live complete graph.

Theorem 1 (vector live-star identity)**.** **[PROVED]** *Every completed history satisfies*

$$D(h) = \sum_{x \text{ opened}} s_{L_x}(x), \tag{2}$$

where L_x is the live set at the instant x opens.

Proof. Fix a real pair xy and suppose x opens first. The vertex y is live when x opens, so xy occurs once in $s_{L_x}(x)$. If the activity intervals overlap, x is still live when y opens and the coordinate occurs a second time in $s_{L_y}(y)$; the two occurrences cancel over $\mathbb{F}_2$. If the intervals are disjoint, x has already closed when y opens and there is no second occurrence. The coordinate of the right-hand side is therefore exactly the coordinate of $D(h)$. $\square$

Corollary 2 (scalar live-degree identity). **[PROVED]** *For every graph G,*

$$F_G(h) = \sum_{x \text{ opened}} \deg_{G[L_x]}(x) \pmod 2.$$

This upgrades the earlier scalar potential calculation to an identity before a graph is chosen. The graph enters only through the final linear functional $z \mapsto \langle A_G, z\rangle$.

3 Affine response is exactly the theorem

Fix which seat seeks odd parity, and fix a deterministic strategy S for that attacker. Move legality is graph-independent, so S can be regarded as a strategy in the schedule game before an adjacency vector is chosen. Let $\mathcal{H}_S$ be the set of terminal histories compatible with S and with arbitrary legal responses by the other player.

Theorem 3 (affine separation). **[PROVED]** *The strategy S forces odd score for some graph on R if and only if*

$$0 \notin \mathrm{Aff}_{\mathbb{F}_2}\{D(h) : h \in \mathcal{H}_S\}.$$

Consequently, the general linking theorem is equivalent to the following graph-free statement for both choices of attacker seat:

$$\textit{for every } S, \qquad 0 \in \mathrm{Aff}_{\mathbb{F}_2}\{D(h) : h \in \mathcal{H}_S\}. \tag{3}$$

Proof. If S forces odd score for a graph with adjacency vector A, then $\langle A, D(h)\rangle = 1$ for every $h \in \mathcal{H}_S$. The whole affine hull therefore lies in the affine hyperplane $\langle A, z\rangle = 1$, which excludes zero.

Conversely, write the affine hull as $a + W$. If it excludes zero, then $a \notin W$. Elementary linear separation over $\mathbb{F}_2$ supplies a linear functional ℓ with $\ell(W) = 0$ and $\ell(a) = 1$. Every linear functional on $\mathcal{E}_R$ is pairing with the adjacency vector of a simple graph on R. For that graph, $\ell(D(h)) = 1$ for every response history, so S forces odd score. □

Equivalently, one must select an odd-cardinality formal multiset of response histories whose disjointness vectors XOR to zero. This is an $\mathbb{F}_2$-flow problem: attacker nodes route all incoming weight along the move prescribed by S, while defender nodes may split it among an odd number of outgoing moves. Uniformly summing all response leaves does *not* work in general; the flow must be selected.

There is a useful exact homogeneous recursion. At a history h, let $Z(h) \subseteq \mathbb{F}_2 \oplus \mathcal{E}_R$ be the span of $(1, D_h(\lambda))$ over response leaves λ below h, where D_h is the future disjointness increment. Give a close of a real front f the charge $c = \sum_{u \in U \cap R} fu$, and give an open, dummy close, or pass charge zero. This sums to $D(h)$ because a pair becomes disjoint exactly when its earlier vertex closes while the later one is untouched. Put

$$\tau_c(a, z) = (a, z + ac).$$

Then a terminal node has $Z = \mathrm{span}\{(1,0)\}$; an attacker node, unary after fixing S, has $Z(h) = \tau_c Z(h')$; and a defender node has

$$Z(h) = \mathrm{span}_m \, \tau_{c_m} Z(h_m).$$

The desired affine response is exactly $(1, 0) \in Z(\text{root})$. Thus the open problem is a label-preserving contraction of this recursion, not an ordinary contraction of the response tree: ordinary boundaries have even augmentation, while the required leaf chain has odd augmentation.

[TESTED] The existing exhaustive graph census through eight real vertices plus dummy, both attacker seats, implies (3) in those dimensions by Theorem 3. Independent graph-free policy searches also found no affine offset through eight real vertices; selected structured policies at nine and ten real vertices likewise had zero offset. These are finite checks, not a proof of (3).

4 A solved parity-cell subclass

The true/false-twin argument in the earlier draft can be weakened substantially.

Theorem 4 (parity-cell pairing)**.** **[PROVED]** *Suppose all vertices, including the dummy if present, admit a perfect partition $\mathcal{M}$ into two-element cells such that*

$$e_G(A, B) = 0 \pmod 2 \qquad \textit{for every distinct } A, B \in \mathcal{M}. \tag{4}$$

Then the second player forces even flip parity.

Proof. When the first player opens one member of a cell A, the second opens its mate. When the first player closes the FIFO-front member of A, its mate is the next queue front and the second closes it. Thus untouched vertices are always a union of whole cells, and the queue is a concatenation of whole cells. Ko causes no exception because the forced second opening completes the first cell.

When the two members a, a' of A close consecutively, their two flip bits sum to

$$\deg_U(a) + \deg_U(a') = \sum_{B \subseteq U} e_G(A, B) = 0$$

by (4). Every flip is contained in one such consecutive pair, so the total parity is even. □

Twin pairs satisfy (4), but the condition only asks for even cross-parity between cells; no automorphism is required.

[TESTED] This theorem is still not general. Among the 1,044 isomorphism classes on seven real vertices, 64 boards obtained by adjoining the isolated dummy have no such partition. The first witness in the local ordering has real edges

$$02, 04, 06, 14, 15, 23.$$

The general linking theorem nevertheless passes exact minimax on every one of those boards. Seidel switching cannot repair the pairing obstruction: between two two-cells A, B, switching a set S toggles

$$|A \cap S||B \setminus S| + |A \setminus S||B \cap S| \equiv ab + ab = 0 \pmod 2.$$

5 FIFO gives a cut-space filtration

Let $L \subseteq R$ be a live real set and let $f \in L$ be the FIFO front. Let $E(L) = \mathbb{F}_2^{\binom{L}{2}}$ and let

$$\mathrm{Cut}(L) = \mathrm{span}\{s_L(x) : x \in L\}.$$

This is the cut space of the complete graph on L.

Theorem 5 (front splitting)**. [PROVED]** *There is a canonical direct sum*

$$E(L) = \mathrm{Cut}(L) \oplus E(L \setminus \{f\}). \tag{5}$$

The projection to the second summand is

$$(T_f z)_{ij} = z_{ij} + z_{fi} + z_{fj}, \qquad i, j \neq f. \tag{6}$$

It kills every live star and is the identity on the canonical $E(L \setminus \{f\})$ summand.

Proof. For $x = f$, the two incident coordinates in (6) cancel; for $x = i \neq f$, the ij and fi coordinates of $s_L(i)$ cancel. Hence T_f kills all generators of $\mathrm{Cut}(L)$. If z is supported away from f, then $T_f z = z$. Finally,

$$\dim \mathrm{Cut}(L) + \dim E(L \setminus \{f\}) = (|L| - 1) + \binom{|L|-1}{2} = \binom{|L|}{2},$$

so the two subspaces are complementary and exhaust $E(L)$. □

There is a front-independent description of the same quotient. Regard $z \in E(L)$ as a simplicial 1-cochain on the full simplex on L, and put

$$(\delta z)_{ijk} = z_{ij} + z_{ik} + z_{jk}.$$

Then $\ker \delta = \mathrm{Cut}(L)$ and

$$(T_f z)_{ij} = (\delta z)_{fij}. \tag{7}$$

Thus T_f is the chart based at f of the odd-triple two-graph of z; graphically it is Seidel complementation at f followed by deletion of f [4, 5]. Equivalently, if B is the alternating form with Gram matrix z and

$$E_0 = \{x \in \mathbb{F}_2^L : \textstyle\sum_i x_i = 0\}, \qquad b_i^{(f)} = e_i + e_f,$$

then $T_f z$ is the Gram matrix of $B|_{E_0}$ in the basis $\{b_i^{(f)} : i \neq f\}$. Changing the basepoint from f to g gives

$$b_f^{(g)} = b_g^{(f)}, \qquad b_i^{(g)} = b_i^{(f)} + b_g^{(f)}.$$

In particular, after restricting away from distinct f, g,

$$T_g(T_f z) = T_g z. \tag{8}$$

This absorption law says that earlier quotient gauges cancel exactly; it does not say that their affine payoff offsets cancel.

While f remains front, opens do not change the live set. By Theorem 1, all their contributions lie in the current $\mathrm{Cut}(L)$ layer. Once FIFO deletes the real front f, every later contribution is supported on $E(L \setminus \{f\})$. Repeating the splitting along the forced closure order gives a triangular decomposition of the complete play score. This is the missing structural use of FIFO: the earlier scalar potential was valid even if any queued vertex could close and therefore could not prove the theorem.

The dummy needs a companion state in this induction. Its star is zero, but closing it does not shrink the real edge space. A proof must use that zero star to adjust affine augmentation and then transfer the odd response flow to the unchanged real live set. Treating the dummy as an ordinary deleted front silently loses precisely the exceptional case.

6 What the cone homotopy does and does not contract

Fix a real front f and a terminal continuation λ. Let $p_f(\lambda) \in \mathbb{F}_2^{L\setminus\{f\}}$ be the indicator of the real vertices still untouched when f closes, and let $W_f(\lambda)$ be the away-f disjointness vector accumulated by the whole continuation. Directly from the definition of disjoint intervals,

$$D_h(\lambda) = \sum_i p_f(\lambda)_i \, fi + W_f(\lambda), \qquad T_f D_h(\lambda) = W_f(\lambda) + \delta p_f(\lambda), \tag{9}$$

where $(\delta p)_{ij} = p_i + p_j$. Hence for every formal response-leaf chain c, of either augmentation,

$$D_h(c) = 0 \quad \Longleftrightarrow \quad p_f(c) = 0 \ \text{ and } \ W_f(c) = 0. \tag{10}$$

The two-graph chart therefore replaces the desired edge moment by a *joint* cut/continuation moment; it does not discard the correlation.

One half of that joint target does contract cleanly at the root.

Lemma 6 (initial cut-moment contraction)**.** **[PROVED]** *Suppose the attacker opens a real initial front f, so the isolated dummy remains untouched and ko forbids the defender's immediate close. For every fixed attacker strategy up to the close of f,*

$$0 \in \mathrm{Aff}\{p_f(\lambda) : \lambda \text{ is a compatible pre-close response}\}.$$

Proof. If zero were absent, separation would give weights $a_i \in \mathbb{F}_2$ on the remaining real vertices such that the attacker forced $a \cdot p_f = 1$. Give the dummy weight zero, and let s be the total weight of the still-unselected vertices. On the ko-forced first response, the defender selects the dummy when $s = 0$, and selects a weight-one real vertex when $s = 1$; either choice leaves $s = 0$. On every later response, close f when $s = 0$, and select a weight-one vertex when $s = 1$. If the attacker selects weight zero, s remains zero and the defender closes. If the attacker selects weight one from a zero total, the number of remaining weight-one vertices was positive and even, so another remains and the defender restores $s = 0$. Whenever the attacker closes f, therefore, $a \cdot p_f = 0$, a contradiction. □

The scalar separation proof has a useful two-round refinement. It repairs the first explicit failure of degree-only pairing, although it still does not iterate through an arbitrary FIFO prefix.

For a finite graph H, put

$$p_v = \deg_H(v) \pmod 2, \qquad b_v = \sum_{u \sim_H v} p_u \pmod 2, \qquad c(v) = (p_v, b_v).$$

Thus b_v is the parity of the odd-degree neighbours of v.

Lemma 7 (two-bit handshake refinement)**.** **[PROVED]** *If H has even order, every one of the four colour classes $c^{-1}(i,j)$ has even cardinality. In particular every vertex has a distinct same-colour mate.*

If $c(v) = c(w)$ and $A = \{v, w\}$, put $r_x = e_H(x, A) \pmod 2$ for $x \notin A$. Then each of the four sets

$$\{x \notin A : (p_x, r_x) = (i,j)\}, \qquad (i,j) \in \mathbb{F}_2^2,$$

also has even cardinality.

Proof. Let $O = \{v : p_v = 1\}$ and $E = V(H) \setminus O$. Handshaking gives $|O| = 0$ in $\mathbb{F}_2$. Moreover

$$\sum_{v \in O} b_v = 2e_H(O) = 0,$$

so both b-fibres in O are even. The cut $e_H(O, E)$ is even, since

$$e_H(O, E) = \sum_{v \in O} \deg_H(v) = |O| = 0$$

in $\mathbb{F}_2$. Hence the $b = 1$ fibre in E is even; $|E|$ is even because $|V(H)|$ and $|O|$ are even, so the remaining fibre is even as well.

For the second assertion, the four parity moments of the map $x \mapsto (p_x, r_x)$ on $V(H) \setminus A$ vanish:

$$\sum 1 = 0, \quad \sum p_x = 0, \quad \sum r_x = p_v + p_w = 0, \quad \sum p_x r_x = b_v + b_w = 0.$$

The internal edge vw, when present, contributes $p_v + p_w = 0$ to the last two displayed identities. The four fibre indicators are the four polynomials $(1+p)(1+r), (1+p)r, p(1+r), pr$ over $\mathbb{F}_2$; their sums therefore vanish. □

Remark 8 (what the second bit buys). Suppose an attacker opens v and the defender opens a same-colour mate w. If the attacker closes v, closing w makes the two flip charges cancel. If the attacker instead opens x, Lemma 7 supplies a y with both $p_y = p_x$ and $e_H(y, A) = e_H(x, A)$. Thus the new cell matches the opening-charge parity and is orthogonal to the first FIFO cell. The equality $b_v = b_w$ is exactly the missing joint moment in the first degree-only counterexample.

This is a two-round statement, not an induction: the new cell can split the four balanced fibres unevenly. A later response may have to close rather than open, which is the correlated continuation obstruction in Question 14.

[TESTED] Exact minimax on every graph isomorphism class through seven real vertices plus the dummy gives a stronger root fact: whenever the same-colour mate of Lemma 7 applies, *every* such reply is winning for the second-seat even player. When the real graph has even order and the attacker opens the unmatched dummy, every real second opening is winning through six real vertices. The first-seat even player also wins by opening the dummy throughout the seven-vertex census. The optional `refinement` stage of `experiments/linking_game.py` reproduces these checks. They support the two-bit prefix but do not prove that its balanced fibres survive later FIFO switches.

7 Scalar response faces and the least counterexample

It is useful to isolate exactly what an induction on a smaller even board would have to prove. Fix the player seeking odd parity and, for any game state h, let $R(h) \subseteq \mathbb{F}_2$ be the set of future charge bits that this attacker can force. If a move of charge c leads to h', write $c + R(h') = \{c + r : r \in R(h')\}$. Backward induction gives

$$R(h) = \begin{cases} \bigcup_m (c_m + R(h_m)), & h \text{ is an attacker node,} \\ \bigcap_m (c_m + R(h_m)), & h \text{ is a defender node,} \end{cases} \qquad R(\text{terminal}) = \{0\}. \tag{11}$$

Here a forced pass is the sole move from its state and has charge zero. For a live vertex set A, use the following canonical roots:

	Q	U	mover / ko
$T(A)$	$()$	A	attacker / false
$K_x(A)$	(x)	$A - \{x\}$	defender / true
$S_x(A)$	(x)	$A - \{x\}$	defender / false
$Q_{xy}(A)$	(x, y)	$A - \{x, y\}$	attacker / false.

The same letters denote their force sets. The boundary values are $T(\varnothing) = K_x(\{x\}) = \{0\}$. Away from the empty-union boundary, directly from (11),

$$T(A) = \bigcup_{x \in A} K_x(A) \quad (A \neq \varnothing), \qquad K_x(A) = \bigcap_{y \neq x} Q_{xy}(A) \quad (|A| \geq 2), \tag{12}$$

and

$$S_x(A) = K_x(A) \cap \big(\deg_{A-\{x\}}(x) + T(A - \{x\})\big). \tag{13}$$

Thus the scalar even-board statement is

$$E(A): \qquad 1 \notin T(A) \quad (|A| \text{ even}). \tag{14}$$

It says that a first-seat odd attacker cannot force odd parity; by finite determinacy, the second player can force even parity.

Lemma 9 (least-counterexample odd routing)**.** **[PROVED]** *Suppose A is a least-order even counterexample to* (14)*. Then there is an $x \in A$ with $1 \in K_x(A)$. Put*

$$Y = \{y \in A - \{x\} : \deg_A(y) = \deg_A(x) \pmod 2\}.$$

The set Y has odd cardinality. For every $y \in Y$, every target-1 strategy witnessing $1 \in Q_{xy}(A)$ begins by opening a third vertex z_y; it cannot begin by closing x.

At the resulting defender state (x, y, z_y), such a strategy must satisfy both

$$1 + \deg_{A-\{x,y,z_y\}}(x) \in Q_{yz_y}(A - \{x\}) \tag{15}$$

and target-1 forceability after every defender opening $w \in A - \{x, y, z_y\}$.

Proof. Direct play proves $E(A)$ for $|A| = 0, 2$: on two vertices the forced opening pair is followed by two zero-charge closes. Hence a least counterexample has order at least four, and Equation (12) supplies x without meeting either boundary case. Each degree-parity class in an even-order graph has even cardinality, so deleting x leaves the odd set Y.

Fix $y \in Y$. If the attacker closes x from $Q_{xy}(A)$, the charge is

$$c = \deg_{A-\{x,y\}}(x),$$

and the child is $S_y(A - \{x\})$. By minimality, $T(A - \{x, y\}) \subseteq \{0\}$, so (13) gives

$$S_y(A - \{x\}) \subseteq \{\deg_{A-\{x,y\}}(y)\}.$$

But equality of the full degree parities gives

$$\deg_{A-\{x,y\}}(x) = \deg_{A-\{x,y\}}(y),$$

where the possible edge xy cancels from both sides. The remaining target after the close is $1+c$, not c, so this branch cannot force target 1. The attacker must therefore open some z_y. The defender may now close x or open any remaining w. Applying (11) to those children gives (15) and the stated universal open conditions. ∎

The odd routing is a real strengthening over the handshake lemma: it couples an odd family of sibling histories. It still cannot be collapsed one sibling at a time. A tempting twisted induction would label $C = A - \{x, y\}$ by $a(u) = B(x + y, u)$ and ask for a w with

$$1 + a(z) + a(w) \notin Q_{zw}(C).$$

[TESTED] This is false first at $|C| = 4$. Take the path $5-2-3-4$, put $(a_2, a_3, a_4, a_5) = (1, 0, 1, 0)$, and $z = 5$. Exact recursion gives

$$T(C) = \{0\}, \qquad Q_{52} = \{0\}, \quad Q_{53} = \{0, 1\}, \quad Q_{54} = \{0\},$$

while the three shifted targets are $0, 1, 0$. All three survive. This is realized inside the six-vertex graph

$$E = \{03, 12, 13, 14, 23, 25, 34\}, \qquad (x, y) = (0, 1).$$

Hence ordinary smaller-board induction, even augmented by one complete cut bit, does not prove Lemma 9's required joint sibling cancellation.

The lemma cannot simply be followed by an independent quotient induction. On real vertices $0, 1, 2$ plus d, fix a second-seat attacker so that the line

$$O_d, O_1, C_d, O_0, C_1, O_2, C_0, C_2$$

is forced below the displayed defender choices. At the C_1 branch from $Q = (1, 0), U = \{2\}$, one has $p_1 = e_2$, $W_1 = 0$, $D = 12$, and $T_1 D = 02$. The subtree below that branch has one leaf. The fictitious 02 terms in $W + \delta p$ cancel only after the cut and quotient pieces are recombined; no compatible continuation chain realizes either term separately. This is the smallest broken commuting diamond behind the false branchwise induction.

8 Why the deterministic local induction fails

The live-degree pairing heuristic is correct only conditionally. At a turn of the block initiator the queue has even length. If the initiator opens x, the responder would like to open an untouched y with the matching live-degree parity. If no such y exists and the old queue is nonempty, closing the old front rotates

$$(a_1, b_1, \dots, a_r, b_r, x) \longmapsto (b_1, a_2, \dots, a_r, b_r, x).$$

Neither operation is a complete strategy.

[TESTED] The smallest useful witnesses are exact minimax states, not counterexamples to the linking theorem.

1. On $K_2 + d$, after

   $$A : O_d, \qquad D : O_0, \qquad A : C_d,$$

   the defender faces $Q = (0)$ and $U = \{1\}$. The winning move is O_1; the apparently natural mirror close C_0 loses.

2. On the real path $3 - 0 - 1 - 2$ plus d, after

   $$A : O_0, \qquad D : O_1, \qquad A : O_d,$$

   the defender faces $Q = (0, 1, d)$ and $U = \{2, 3\}$. Both available opens have the wrong live-score parity, and closing 0 loses. The unique winning move is the proactive wrong-parity open O_3, which creates an even odd-front corridor and stores a debt before the local trap appears.

Thus a state descriptor consisting only of current debt, front parity, parity classes in U, and one FIFO rotation is insufficient. The missing datum is a recursive repair certificate in the untouched graph. This is exactly what the affine flow is allowed to encode globally.

9 The exact local residue

The scalar cut potential gives a complete local classification, provided its limitation is stated explicitly. Regard the queue as a set when counting edges and put

$$P(Q,U) = e_G(Q,U) \pmod 2.$$

For a move m, let $\chi(m)$ be the change in P. If $R = Q \cup U$, direct edge accounting gives

$$\chi(O_x) = \deg_{G[R]}(x), \qquad \chi(C_f) = \deg_U(f), \qquad \chi(\text{pass}) = 0.$$

The middle term is the actual flip.

Proposition 10 (charge-matching and the poison state)**.** **[PROVED]** *Suppose $P = 0$, $|U|$ is even, and the position is nonterminal. After an opponent open, there is always a distinct legal open with the same χ. If a reply is due after an opponent close of charge c, there is a legal response of charge c except in exactly the following case:*

$$\begin{aligned} c = 0, \qquad U \neq \varnothing, \qquad \deg_{R'}(u) = 1 \pmod 2 \quad \textit{for every } u \in U, \\ Q' = \varnothing \quad \textit{or} \quad \deg_U(\text{front}\, Q') = 1 \pmod 2. \end{aligned} \tag{16}$$

where $Q' = Q \setminus \{f\}$ and $R' = Q' \cup U$ after the close.

Proof. At $P = 0$,

$$\sum_{u \in U} \deg_R(u) = e(U,Q) = 0 \pmod 2,$$

because edges internal to U are counted twice. Hence the odd-charge opens form an even set; since $|U|$ is even, the even-charge opens do too. The charge class containing the opponent's opener therefore contains another vertex. Opening never has a ko obstruction, and the live set R is unchanged by an open, so this is a legal matching response.

If the opponent closes f with charge c, then

$$\sum_{u \in U} \deg_{R'}(u) = e(U,Q') = c \pmod 2.$$

For $c = 1$ an odd-charge open exists. For $c = 0$ an even-charge open exists unless every untouched vertex has odd charge; in that exceptional case a matching response still exists precisely when the new FIFO front exists and has even degree into U. A close clears ko, so that front close is legal. This is exactly the complement of (16). When U is empty all remaining closes, including the close after the unique terminal pass, have charge zero. □

Matching χ restores $P = 0$, but does *not* by itself pair the flip score: a mixed odd-close/odd-open round contains one actual flip. This is the debt which the global chain must transport. The dummy prevents (16) while it is untouched (it is an even-charge open) or is the new front (it is an even close), but it may be buried in the queue or already closed. The proposition therefore isolates the local obstruction without pretending to solve its recurrence.

One portion of that recurrence is nevertheless well founded. Let g be the accumulated flip debt and keep $P = e_G(Q,U)$ as above.

Lemma 11 (fixed-front poison descent). **[PROVED]** *Suppose it is the defender's turn, $g = 1$, and the front f has positive even degree into U; ko may be set or clear. The defender can open a neighbour of f so that either the debt is discharged within the next attacker–defender round or the degree drops by exactly two. If the result is still positive the lemma may be repeated; if it is zero, the descended anticomplete front can be closed but need not discharge the debt. Hence the fixed-front loop terminates. Moreover*

$$H = g + P$$

is unchanged by every close. Thus $g = 1, H = 0$ excludes a singleton anticomplete front, although H is not generally preserved by opens.

Proof. Opening one neighbour changes $\deg_U(f)$ from even to odd and clears ko, since the queue was already nonempty. If the attacker closes f, that odd close discharges the debt. If the attacker opens a nonneighbour, the front remains odd and the defender closes it, again discharging the debt. The only remaining attacker move is to open a second neighbour; then the front is even again, the defender faces the same local problem when its degree remains positive, and exactly two of its untouched neighbours have been removed. At degree zero the neighbour-open loop stops; ko is clear at this descended boundary, so the even close is legal and carries no flip.

A close of charge c changes both g and P by c, so it fixes H. If $g = 1$ and $H = 0$, then $P = 1$, whereas a singleton queue whose front is anticomplete to U has $P = 0$. Finally an open x changes H by $\deg_{G[Q\cup U]}(x)$, which may be odd; therefore this exclusion is a local corridor condition, not a global invariant. □

The first obstruction to same-action pairing does have a complete local repair. Write $B(u, v)$ for adjacency in the current graph.

Lemma 12 (two-switch tail). **[PROVED]** *Suppose the defender moves just after an attacker close, the score debt is zero, and*

$$U = \{x, y\}, \qquad Q = (v_1, \ldots, v_{2r+1}).$$

If

$$B(v_{2i-1}, y) = B(v_{2i}, y) \quad (1 \le i \le r), \qquad B(v_{2r+1}, y) = B(x, y), \tag{17}$$

then the defender forces even final score with at most two changes of action type relative to the attacker's preceding move.

Proof. The defender first opens x, so the even queue is paired as $(v_1, v_2), \ldots, (v_{2r+1}, x)$ and every pair has equal adjacency to the sole untouched vertex y. While the attacker closes, the defender closes the next front; the two flip bits in each pair cancel by (17). If the attacker opens y, the defender closes the front. Now U is empty and every later close has charge zero. If the attacker never opens y before the pairs are exhausted, y is the harmless terminal singleton. The initial close/open and the possible later open/close are the two switches. □

[TESTED] Same-action adaptive pairing first fails, without a dummy, on two of the seven exceptional order-eight graphs: `GCZN^{` and `GEjt~{`. Call them G_1, G_2 in the table below. Exact minimax gives minimum worst-case switch count two on each. Their worst paths enter Lemma 12 at

	Q	U	(x, y)
G_1	$(1, 2, 4, 3, 5)$	$\{6, 7\}$	$(6, 7)$
G_2	$(1, 2, 3, 4, 5)$	$\{6, 7\}$	$(6, 7)$.

In both graphs 7 dominates the remaining live graph, so (17) is immediate. This proves the tail repair, not the still-missing prefix theorem that would force such a tail on every board.

10 The zero-normalized kernel and dynamic queue matchings

The order-eight search reveals a stronger scalar target than (14). Sample play after every defender move, with the odd-seeking player about to move and accumulated flip score zero. Away from the unique terminal-pass boundary, the queue then has even length. Put

$$\rho(U, Q) = |U| + \frac{|Q|}{2};$$

every two-move block decreases ρ by one.

Proposition 13 (zero-normalized repair recursion)**.** **[PROVED]** *Let Q have even length. There is a strategy returning the flip score to zero after every defender move if and only if the following recursive predicate $\mathsf{Safe}(U, Q)$ holds.*

- $\mathsf{Safe}(\varnothing, ())$ *holds.*
- $\mathsf{Safe}(\{z\}, ())$ *holds as the terminal macro O_z, forced zero-charge pass, C_z. The pass is the defender's move; after it the score is still zero, and the last close also has charge zero.*
- *After an attacker open O_z, a legal zero-charge reply is either O_w for some $w \in U - \{z\}$, leading to*
$$\mathsf{Safe}(U - \{z, w\}, Qzw),$$
or, when $Q = (f, Q_0)$ and $\deg_{U-\{z\}}(f) = 0$, the close C_f, leading to
$$\mathsf{Safe}(U - \{z\}, Q_0 z).$$
- *After an attacker close C_f of charge $\epsilon = \deg_U(f)$ from $Q = (f, h, Q_0)$, a same-charge reply is C_h when $\deg_U(h) = \epsilon$, leading to $\mathsf{Safe}(U, Q_0)$. When $\epsilon = 0$, the defender may instead open $w \in U$, leading to $\mathsf{Safe}(U - \{w\}, hQ_0w)$.*

Except inside the displayed terminal macro, every legal attacker move must have at least one listed reply whose child is safe.

Proof. All opens and passes have flip charge zero, and the only nonzero move charge is the displayed untouched degree of a FIFO front. Consequently the list contains exactly the replies whose charge equals the preceding attacker charge. The singleton macro is the only time the defender has no ordinary reply: opening the last untouched vertex onto an empty queue creates ko, so the forced pass clears it and leaves one final zero-charge close. Every other child has rank $\rho - 1$. Induction on ρ, treating the macro as a terminal leaf, proves sufficiency by selecting the recorded child after every attacker move. Conversely, record the first reply of any strategy which restores zero after each defender move, and apply the same argument to every reachable child. This produces the recursion. □

The resulting certificate is a ranked, branch-dependent repair forest. It is not yet a structural existence proof: defining the forest by the winning strategy would be circular. It does, however, state exactly what the finite search certifies and prevents weaker local invariants from being mistaken for the theorem.

[TESTED] Every even-order graph through order eight has $\mathsf{Safe}(V, ())$. On all 12,346 order-eight isomorphism classes, the second player can therefore keep the *actual* flip score zero after every one of her moves. The first failures of the stronger same-action rule occur on `GCZN^{` and `GEjt~{`;

the zero-normalized strategy survives by using a close/open or open/close rotation. Static selectors fail more sharply: on `G?bFf[` an opener's unique same-(p, b) mate is unsafe, and on `GCz^vw` an odd-degree opener's only safe reply has even degree. These are failures of proposed selectors, not of Proposition 13 or the linking theorem.

The rotations have an exact ordered-matching interpretation. At a checkpoint write

$$Q = (v_0, v_1, \dots, v_{2r-1})$$

and match consecutive queue vertices. Pair opens extend this matching and pair closes delete its front edge. If the attacker appends $z = v_{2r}$ and the defender gives the legal zero close C_{v_0}, the new matching is shifted by one place. Let

$$P(W) = \sum_{i=0}^{2r-1} [v_i, v_{i+1}], \qquad W = (v_0, \dots, v_{2r}).$$

Then the old and shifted matching chains differ by $P(W)$ and

$$\partial P(W) = e_{v_0} + e_z.$$

More precisely, with the triangle fan

$$F(W) = \sum_{i=1}^{2r-1} [v_0, v_i, v_{i+1}],$$

one has

$$P(W) + [v_0, z] = \partial F(W). \tag{18}$$

For the adjacency 1-cochain B, therefore,

$$\langle B, P(W) \rangle = B(v_0, z) + \sum_{i=1}^{2r-1} \delta B(v_0, v_i, v_{i+1}). \tag{19}$$

The second term is genuine two-graph holonomy. Endpoint exchange, a queue matching span, or a symplectic basis invariant sees only $B(v_0, z)$ and is therefore insufficient on an arbitrary graph.

The triangle shears are coherent but not flat. If $\tau(i, j, k) = \delta B(i, j, k)$, then

$$\tau(j, k, l) + \tau(i, k, l) + \tau(i, j, l) + \tau(i, j, k) = 0, \tag{20}$$

which is $\delta^2 B = 0$. Thus the ambient complete simplex fills the fan in (18) consistently. The adversarial obstruction is in its coefficients: after fixing an attacker strategy, compatible histories are not closed under deleting an interior letter. Defender coefficients may split, but an attacker coefficient must follow one prescribed child. Projection to compatible histories consequently fails to commute with the simplicial boundary.

This is not repaired by ordinary topology of injective words. Those words form a Boolean cell complex with

$$\partial[v_0 \cdots v_k] = \sum_i [v_0 \cdots \widehat{v_i} \cdots v_k] \quad \text{over } \mathbb{F}_2$$

[6, 7]. Define the strategy carrier here as the Boolean subcomplex generated by the attacker-compatible pre-close tail words. Even with a dummy, attacker pruning need not leave this carrier acyclic. On $K_2 + d$, let the attacker open a real front f and, after the forced-open reply is chosen as

O_a or O_d, close f immediately. Before the first close the carrier consists of the two isolated vertices $[a]$ and $[d]$; the joining cells $[ad]$ and $[da]$ have been pruned. Its reduced H_0 is nonzero. The linking response still exists because the two close charges differ. Hence any successful carrier theorem must retain the charge and continuation labels; ordinary shellability or a constant-coefficient cone is not enough.

Three further exact obstructions delimit the remaining Boolean contraction. First, the two-bit colour is a prefix moment, not a winning selector. **[TESTED]** On the eight-vertex graph

$$\begin{aligned} E = \{&01, 03, 05, 07, 12, 14, 15, 17, 24, 26, 27, 34,\\ &37, 45, 46, 47, 57, 67\}, \end{aligned}$$

vertices $6, 7$ have the same (p, b) colour, but Q_{67} is unsafe while Q_{76} is safe. The unique first spoiler in Q_{67} is O_5. There is also an opener with no safe same-colour mate: in graph6 class `G?ben[`, with

$$E = \{04, 05, 15, 25, 06, 16, 36, 56, 07, 17, 27, 47, 57, 67\},$$

vertex 2 has sole same-colour mate 7, yet Q_{27} is unsafe. Its unique first spoiler is O_3; the reverse Q_{72} is safe. Thus neither orientation nor existential choice within one colour class proves the root theorem.

Second, a safe residual strategy cannot always be lifted by insisting on an open response. On

$$\begin{aligned} E = \{&01, 02, 03, 04, 05, 06, 12, 13, 14, 17, 23, 26,\\ &27, 35, 37, 56, 57\}, \end{aligned}$$

the checkpoint

$$Q = (4, 3), \qquad U = \{0, 1, 2, 5, 6, 7\}$$

is safe in the zero-normalized recursion. After attacker O_6, the unique safe reply is the zero close C_4, leading to $Q = (3, 6)$ and $U = \{0, 1, 2, 5, 7\}$. The five open replies by $0, 1, 2, 5, 7$ are all unsafe; respective short spoilers may be chosen as C_4, C_4, O_7, O_7, O_1. Hence the phase-changing OC branch is not an artifact of the finite search: it is indispensable even when $|U|$ was even.

Finally, time reversal does not repair the strategic pruning. For a pass-free complete event word put

$$\rho(e_1 \cdots e_m) = \bar{e}_m \cdots \bar{e}_1, \qquad \overline{O_v} = C_v, \quad \overline{C_v} = O_v.$$

[PROVED] This is a legal FIFO schedule, swaps the two seats, and preserves D coordinatewise. Indeed, if both opening and closing order in the original word are $v_1, \ldots, v_n$, both orders after reversal are $v_n, \ldots, v_1$, so FIFO is preserved. A pass-free word has no adjacent singleton block $O_v C_v$, hence reversal creates no ko violation. Position i moves to $2n + 1 - i$, which swaps seat parity, while each activity interval $[o_v, c_v]$ reflects to $[2n + 1 - c_v, 2n + 1 - o_v]$; interval disjointness, and therefore D, is unchanged.

This identity does not extend to a strategy contraction. If one attempts to set $\rho(P) = P$, a terminal singleton component O_x, P, C_x moves to the beginning; when other vertices remain untouched there, its pass is not forced and is illegal. Even without passes, fixed strategies are anti-causal under ρ. Put the original attacker in the even positions. One deterministic attacker strategy contains both histories below: after O_0 it prescribes O_d; after the defender's choice C_0 or O_1 it prescribes respectively O_1 or C_0; and after C_d it prescribes C_1. The smallest such local witness has real vertices $0, 1$ plus d:

$$\begin{array}{r|c} h_C = O_0 O_d C_0 O_1 C_d C_1 & D(h_C) = 01, \\ h_O = O_0 O_d O_1 C_0 C_d C_1 & D(h_O) = 0. \end{array}$$

The two histories reconverge across the commuting C_0/O_1 diamond, whose label difference is the edge coordinate 01. Reversal gives a common prefix $O_1 O_d$ but demands different moves, C_1 and O_0, at the same odd-position attacker node. Hence the reversed pair cannot lie under one deterministic reversed attacker strategy. Separately, the diamond involution ι exchanging C_0 and O_1 swaps h_C and h_O, so

$$\mathbb{F}_2\{h_C, h_O\}^{\iota} = \{0, h_C + h_O\}.$$

Both invariant chains have even augmentation (and $D(h_C + h_O) = 01$), whereas the valid odd zero-moment certificate is the nonsymmetric singleton h_O. Diamond symmetrization therefore destroys the odd augmentation rather than proving it.

11 The remaining global lemma

The proof target can now be stated without Gold forms, graph enumeration, or a particular local menu.

Question 14 (global FIFO affine contraction). **[OPEN]** Fix either attacker seat and a deterministic attacker strategy S in the schedule game with one isolated dummy. Does there always exist an odd $\mathbb{F}_2$-flow through the defender branches whose terminal disjointness moment is zero, with the flow allowed to couple different defender children and different levels of the front filtration?

An affirmative answer proves the general linking theorem by Theorems 3 and 5. The tempting stronger claim—that projected continuation hulls of different defender children meet one common coset—is false. At the $P_4 + d$ state above, fix a history-dependent attacker which follows different continuations after O_2 and O_3. Under the front projection T_0 to the edge coordinates $12, 13, 23$, exact recursion gives the two open-child continuation hulls

$$\{0\}, \qquad \mathrm{Aff}\{23, 12 + 23\}.$$

They are disjoint. Exhaustive recursion over the smaller states with two real open children shows that this is the first such childwise failure.

Thus a valid contraction must transport nonzero target cosets and cancel them *across* defender children; it cannot solve every projected child as an independent zero-target subproblem. In particular, an induction classified only by dummy presence, live-set parity, attacker seat, and front position is false on reachable states. The local relations

$$s_L(d) = 0, \qquad \sum_{x \in L \cap R} s_L(x) = 0$$

provide the cone and the cut dependence, but not the required global coupling.

This boundary also explains why the nearest imported techniques do not close the problem. A standard strategy-stealing pass must be optional and harmless; the dummy has two mandatory events which change FIFO legality. A static copycat argument requires an involution or a parity-cell partition, absent on the 64 finite witnesses above. Local complementation changes the required pairing data but is not a legal FIFO move. Queue layouts supply terminology for nonnesting but no adversarial parity strategy.

[TESTED] A useful stronger conjecture is false: without the dummy, the initial mover does not always force even score on even-order graphs. The first failures occur at order eight, in exactly seven isomorphism classes: `GCZN^{`, `GCrU~{`, `GEhvn{`, `GCx}~{`, `GEjt~{`, `GEl~v{`, and `GUZv^{`. They are anti-mover-controlled: the second player can force the target parity, whether that target is even or odd. In particular the weaker finite conjecture that a second-seat even player wins every even-order no-dummy board survives this screen, but it is not proved. Bare alternating rank or Witt class cannot classify these outcomes: P_3 and $K_2 \sqcup K_1$ have congruent rank-two alternating adjacency forms over $\mathbb{F}_2$ but different seat outcomes.

[PROVED] The smallest genuine affine cancellation is already a three-history fan, not a dummy cone. With real vertices $1, 2, 3$, let a first-seat attacker open the dummy and thereafter close whenever possible, otherwise opening the least untouched vertex. After the shared prefix O_d, O_1, C_d, the defender can realize the three compatible terminal histories

	remaining moves	$D(h)$
(i)	O_2, C_1, O_3, C_2, C_3	$\{13\}$
(ii)	O_3, C_1, O_2, C_3, C_2	$\{12\}$
(iii)	C_1, O_2, O_3, C_2, C_3	$\{12, 13\}$.

Their three vectors XOR to zero, although none is zero. In (iii), O_2 opens an empty queue and O_3 is the ko-forced response, so all three rows are legal. The cancellation happens after the dummy has closed.

[TESTED] Three histories do not suffice uniformly. With five real vertices plus d, let a first-seat attacker close whenever legal and otherwise open the least untouched vertex. Its 132 distinct response scores contain neither zero nor three vectors with XOR zero. The minimum odd certificate has support five; one is

$$\begin{aligned} &\{01, 02, 03, 13, 14, 24\}, && \{01, 02, 04, 12, 13, 23, 24\}, \\ &\{01, 03, 04, 12, 14, 23\}, && \{02, 03, 04, 12, 14, 23\}, \\ &\{01, 02, 03, 04, 12, 14, 23\}. \end{aligned}$$

The five rows XOR to zero. Thus the triangular fan is a real local cancellation, not a bounded global normal form for affine response flows.

[TESTED] Even the natural fan over *all* second openings is too rigid. On real vertices $0, 1, 2$ plus d, let the attacker first play O_0; after replies O_1 or O_2 use "open least untouched, otherwise close", and after O_d use "close whenever possible, otherwise open least". Exact terminal affine hulls conditioned on the second reply, in coordinates $01, 02, 12$, are

$$\mathcal{A}_1 = \{0\}, \qquad \mathcal{A}_2 = \{0\}, \qquad \mathcal{A}_d = \{3, 7\}.$$

The immediate star charges of the three branches XOR to zero, but no one-leaf-per-branch transversal does. The global affine target survives by retaining only an odd subflow inside $\mathcal{A}_1$ or $\mathcal{A}_2$. Hence a proof must choose its odd subset of defender branches using continuation data; neither dummy-edge contraction nor the full root fan is a valid fixed recipe.

12 Consequence for the Gold realizer

[PROVED] The reductions in `writeups/goldarf.tex` identify the terminal σ-charge with the overlap parity of G-edges. Since overlap and disjointness partition $E(G)$,

$$\sigma(h) = |E(G)| + F_G(h) \pmod 2.$$

Therefore an affirmative answer to Question 14 would force $\sigma = |E(G)|$ for every support graph and both seats. On a Gold support graph this is the required quadratic value, upgrading the verified $m \in \{4, 8\}$ realizer to every m.

[OPEN] This would still solve only the mechanism half of the original `tis` problem. The realizer is σ-valued; recasting its forced charge as a natural normal-, misère-, or loopy-outcome P-set remains a separate open step.

13 Verification boundary

The following claims have different status and should not be flattened.

- **[PROVED]** The interval/disjointness reduction, vector live-star identity, affine separation equivalence, homogeneous response recursion, parity-cell theorem, front cut-space splitting and two-graph absorption, correlated-moment identity, initial cut contraction, charge/poison classification, two-switch tail, scalar force-set recursion, least-counterexample odd routing, zero-normalized repair recursion, queue-matching phase-fan identity, tetrahedral coherence, and displayed triangular fan and fixed-front poison descent, together with the pass-free reversal identity and its local causal obstruction, are complete proofs in this note.
- **[TESTED]** Exact minimax verifies the general linking theorem on every isomorphism class through eight real vertices plus dummy, both seats, and verifies the two local-strategy witnesses. The parity-cell census is complete through seven real vertices.
- **[TESTED]** The childwise common-coset strengthening is false, first at the displayed $P_4 + d$ state. This is a failure of the proposed induction, not of the linking theorem.
- **[TESTED]** The mandatory full second-opening fan also fails at three real vertices plus dummy, despite the immediate star identity. The global affine target survives inside a single reply branch.
- **[TESTED]** Same-action adaptive pairing first fails on two order-eight no-dummy boards and needs exactly two switches there. A three-leaf affine certificate first fails for the displayed five-real-vertex strategy, whose exact minimum support is five. These are failures of stronger proof templates, not of the linking theorem.
- **[TESTED]** Every even-order graph through order eight has a ranked zero-normalized repair forest. Static degree, two-bit, matching, symplectic-flag, and one-cut selectors have the explicit failures stated above. The finite forests do not prove their own existence at arbitrary order.
- **[TESTED]** Ordered same-colour safety and the weaker existence of a safe same-colour mate both fail at order eight. The open-completion lift also fails at the displayed safe checkpoint, where a zero close is the unique safe response. These are failures of structural proof templates, not counterexamples to the root theorem.
- **[OPEN]** The global FIFO affine-contraction statement (3) is not proved. Neither the complete finite census nor the graph-free affine searches are evidence of a general chain contraction.

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